Fractional powers

1 Express the following in exponential form:

a $\sqrt{x}$ **b** $\sqrt[6]{x}$ **c** $\sqrt[5]{x^7}$ **d** $\sqrt[5]{5}$

e $\sqrt[7]{71}$ **f** $\left(\sqrt[3]{5}\right)^5$ **g** $\sqrt[3]{x^{6a}}$

2 Express $\dfrac{1}{\sqrt{x}}$ with a negative fractional index.

3 Express the following in radical form:

a $3^{\frac{1}{2}}$ **b** $2^{\frac{4}{7}}$

4 Without using a calculator, evaluate the following:

a $121^{\frac{1}{2}}$ **b** $1^{\frac{1}{10}}$ **c** $4^{\frac{3}{2}}$ **d** $(-64)^{\frac{1}{3}}$

e $64^{-\frac{1}{6}}$ **f** $\left(64^{\frac{1}{9}}\right)^{\frac{9}{2}}$ **g** $512^{\frac{1}{3}}$ **h** $625^{\frac{1}{4}}$

i $1024^{\frac{1}{5}}$ **j** $128^{\frac{1}{7}}$ **k** $36^{\frac{3}{2}}$ **l** $10\,000^{\frac{3}{4}}$

m $32^{\frac{2}{5}}$ **n** $64^{\frac{5}{6}}$ **o** $0^{\frac{8}{5}}$ **p** $(-64)^{\frac{4}{3}}$

q $32^{-\frac{2}{5}}$ **r** $(-243)^{-\frac{2}{5}}$ **s** $\left(4a^4\right)^{\frac{1}{2}}$ **t** $8^{\frac{5}{3}}$

5 Simplify the following:

a $\left(\dfrac{3^3}{512}\right)^{\frac{1}{3}}$ **b** $\left(\dfrac{25}{36}\right)^{\frac{1}{2}}$ **c** $\left(\dfrac{169}{16}\right)^{\frac{1}{2}}$ **d** $\left(\dfrac{512}{729}\right)^{\frac{1}{3}}$

e $\left(\dfrac{625}{81}\right)^{\frac{1}{4}}$ **f** $\left(\dfrac{4}{9}\right)^{\frac{5}{2}}$ **g** $\left(\dfrac{64}{27}\right)^{-\frac{1}{3}}$ **h** $\dfrac{1}{243^{-\frac{3}{5}}}$

i $\dfrac{2^{\frac{14}{3}}}{2^{\frac{2}{3}}}$ **j** $\dfrac{2^{\frac{4}{9}}}{2^{\frac{-7}{9}}}$ **k** $\left(\dfrac{x^{15}}{1024}\right)^{\frac{2}{5}}$

6 Simplify the following expressions and then evaluate:

a $36^{\frac{3}{2}} - 32^{\frac{7}{5}}$ **b** $8^{-\frac{4}{7}} \times 8^{-\frac{2}{7}}$ **c** $100^{-\frac{3}{4}} \times 100^{\frac{1}{4}}$ **d** $7^{\frac{1}{7}} \times 49^{\frac{4}{14}}$

7 Find the missing value in the following equation:

$$625^{\frac{3}{\square}} = 125$$

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8 Simplify the following expressions, giving your answers in exponential form:

a $\left(625u^{16}v^{12}\right)^{\frac{1}{2}}$ **b** $y^3 \times \sqrt[3]{y}$ **c** $y^{\frac{6}{5}} \times y^2$ **d** $2y^{\frac{3}{5}} \times 2y^{\frac{2}{5}}$

e $4b^{\frac{3}{2}} \div 2b^{\frac{2}{3}}$

9 Simplify $\sqrt{2} \times \sqrt[3]{2}$ and give your answer in radical form.

10 To evaluate $81^{\frac{3}{2}}$ would it be more efficient to use the property $a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$, or the property $a^{\frac{m}{n}} = \sqrt[n]{a^m}$. Explain your answer.

11 Simplify the following expressions, giving your answers in radical form:

a $\sqrt[8]{x^2}$ **b** $\sqrt[5]{\sqrt{x}}$

12 Simplify $\left(\sqrt[4]{x^5y^3}\right)^{24}$. Assume x and y represent positive numbers.

13 Consider the expression $p^{\frac{q}{r}}$. Complete the following statements:

a One way of understanding this expression is to say that we are raising p to the power of $\square$, then taking the $\square$th root of the result.

b Another way of understanding this expression is to say that we are taking the $\square$th root of p, and then raising the result to the power of $\square$.

Applications

14 The approximate radius, r, of a circle with an area, A, is $r = 0.32A^{\frac{1}{2}}$. Find the approximate radius of a circle with an area of $64\ \text{in}^2$.

15 The surface area, A, of a cube with a volume, V, is $A = 6V^{\frac{2}{3}}$. Find the surface area of a cube with a volume of $1000\ \text{in}^2$.

16 The volume, V, of a cube with a surface area, A, is $V = \left(\frac{A}{6}\right)^{\frac{3}{2}}$. Find the volume of a cube with a surface area of $54\ \text{in}^2$.

17 A company's net profit (in dollars) can be modeled by the equation $P = 160n^{\frac{3}{2}} - 1100$, where n is the number of units sold. What will its net profit be if it manages to sell 16 units?

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